

BUSINESS PLAN

NetPath

Interactive Network Education Platform

Learn networking by building it.

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1. Executive Summary

1.1 Business Concept

NetPath is a web-based, sandbox-driven interactive tutorial platform that teaches computer networking fundamentals through hands-on simulation. Modelled on the learn-by-doing success of LearnGitBranching, NetPath guides learners from the basics of Local Area Networks (LANs) all the way through Wide Area Networks (WANs) by introducing networking devices one at a time, showing how each fits into a progressively more complex topology, and ultimately inviting users to design their own networks via drag-and-drop interfaces.

The platform deliberately abstracts away low-level packet mechanics — transmission times, round-trip delays, dropped packets — so that learners can focus entirely on topology, device roles, and connectivity logic. A lightweight simulation engine confirms whether a user's network is correctly configured and visualises the path a request takes through it, providing immediate, satisfying feedback without overwhelming beginners.

1.2 Target Market

NetPath addresses three overlapping audiences:

- Students in secondary and post-secondary computer science or IT programmes who need an accessible first encounter with networking concepts.
- Self-taught developers and career-switchers preparing for certifications (CompTIA Network+, Cisco CCNA) or roles that require networking literacy.
- Corporate training and onboarding teams seeking a cost-effective, engaging way to bring non-specialist employees up to speed on internal infrastructure concepts.

The global e-learning market was valued at over USD 250 billion in 2024 and is projected to exceed USD 600 billion by 2030, driven by demand for technical upskilling. The networking education sub-segment benefits additionally from the explosive growth of cloud infrastructure roles, cybersecurity hiring, and the proliferation of home-lab culture.

1.3 Proposed Solution

NetPath delivers a structured, progressive curriculum through three interlocking product layers:

- Guided Tutorial Stages — bite-sized lessons that introduce one device or concept (hub, switch, router, DNS server, firewall, NAT gateway, etc.), explain its function with annotated diagrams, then drop the learner into an interactive sandbox to observe its behaviour within a pre-built network.
- Sandbox Design Challenges — at milestone stages, learners are given a set of constraints (e.g. "connect three office floors so that any host can reach the internet") and

must assemble the correct topology by dragging devices onto a canvas and drawing wired or wireless links between them.

- Request Simulation — once a topology is validated, users can fire a simulated HTTP request and watch it hop device-by-device through the network to its destination and back, reinforcing conceptual understanding through visual storytelling rather than numerical metrics.

1.4 Team

NetPath is founded by a small cross-functional team combining deep technical expertise with proven product and education experience:

Role	Background
Co-Founder & CEO	5+ years in EdTech product management; previously led growth at a coding bootcamp platform serving 40,000 learners.
Co-Founder & CTO	Software engineer with expertise in graph-based data structures, React/Canvas rendering, and real-time web apps; open-source contributor to networking simulation tools.
Lead Curriculum Designer	Former university networking lecturer; author of two CompTIA study guides; experience adapting technical content for diverse learner levels.
Head of Design (UX/UI)	Specialist in interactive data visualisation and educational interface design; previously at a major online learning platform.

2. Product Overview

2.1 Core Features

Curriculum Structure

The curriculum is divided into two macro-arcs mirroring real-world networking complexity:

- Arc 1 — Local Area Networks: Ethernet basics, hubs vs. switches, MAC addressing, VLANs, wireless access points, DHCP, and simple LAN design.
- Arc 2 — Wide Area Networks & the Internet: Routers, IP addressing, subnetting, NAT, DNS, firewalls, ISP connectivity, and basic cloud egress paths.

Each arc is broken into short stages (5–10 minutes each). Every stage follows the same pedagogical loop: explain, demonstrate, practise, validate. Learners earn stage badges and unlock the next module only after successfully completing the interactive task, creating a natural sense of progression.

Interactive Sandbox

The drag-and-drop canvas is the heart of NetPath. Key interactions include:

- Device palette — a sidebar of unlocked devices (host, hub, switch, router, access point, DNS server, firewall, NAT box) that can be dragged onto the canvas.
- Connection drawing — click-and-drag between device ports to create wired (Ethernet) links; a toggle switches to wireless association mode for Wi-Fi connections.
- Configuration panel — clicking a device opens a lightweight panel to assign IP addresses, set gateway pointers, or toggle DHCP, keeping technical depth proportional to the current lesson stage.
- Real-time validation — the engine continuously checks the topology against the stage's success criteria and provides colour-coded feedback (green = valid path exists, red = connectivity broken, amber = partial).
- Undo/redo and reset — full history stack so learners can experiment freely without fear of breaking their work permanently.

Request Simulation

Once a network passes validation, users can click "Send Request" to trigger an animated packet journey. A coloured dot travels along the drawn links, pausing briefly at each device with a tooltip explaining the forwarding decision (e.g. "Router checks routing table → forwards to next hop"). The response travels back along the reverse path. No numerical metrics are displayed; the goal is conceptual clarity, not performance modelling.

Progression & Gamification

- Stage-unlock system with illustrated device cards awarded upon completion.
- A topology "portfolio" where learners save and revisit their best sandbox designs.
- Optional timed challenge modes for advanced learners seeking certification prep.
- Leaderboards and peer-sharing for classroom and corporate cohort contexts.

Accessibility & Inclusivity

NetPath is designed to be accessible from day one: full keyboard navigation, screen-reader-compatible SVG canvas labels, colourblind-safe palettes, and support for reduced-motion preferences. The platform is available in-browser with no installation required, targeting both desktop and tablet form factors.

2.2 Technology Stack

Layer	Technology	Rationale
Frontend Framework	React 18 + TypeScript	Component-driven UI; strong ecosystem for complex interactive state.
Canvas / Graph Rendering	React Flow (or custom D3 + SVG layer)	Proven library for node-edge diagrams; supports drag, zoom, and custom node types. Custom SVG layer used for fine-grained animation control.
State Management	Zustand	Lightweight, boilerplate-free global state ideal for canvas + curriculum state.
Network Validation Engine	Custom TypeScript (graph traversal)	BFS/DFS on the topology graph to check reachability and routing correctness. Runs entirely client-side for zero-latency feedback.
Simulation Animator	CSS transitions + requestAnimationFrame	Smooth dot-along-path animation without heavy animation libraries.
Backend / API	Node.js + Express (or Next.js API routes)	Lightweight REST API for auth, progress persistence, and curriculum delivery.
Database	PostgreSQL (primary) + Redis (sessions/cache)	Relational store for user progress; Redis for fast session lookup.
Authentication	Auth0 / NextAuth	SSO, social login, and enterprise SAML support out of the box.
Hosting	Vercel (frontend) + Railway or Render (backend)	Scalable, developer-friendly PaaS with generous free tiers for early growth.
CI/CD	GitHub Actions	Automated testing and deployment on every merge to main.

3. Market Analysis

3.1 Industry Overview

The global e-learning market reached an estimated USD 250 billion in 2024 and is forecast to grow at a compound annual growth rate (CAGR) of approximately 14% through 2030, driven by:

- Accelerating demand for technology skills as cloud adoption, cybersecurity threats, and AI-driven infrastructure transform every industry.
- The normalisation of remote and hybrid work, which has embedded self-directed online learning into mainstream professional development.
- Institutional recognition of micro-credentials and digital badges as valid hiring signals, increasing the value of platform-based certifications.
- Growing "learn-by-doing" expectations among Gen Z learners who find passive video content insufficient and seek interactive, game-like experiences.

Within this macro trend, technical networking education is a particularly underserved niche. The dominant incumbents (Cisco Networking Academy, CompTIA CertMaster, CBT Nuggets) are strong on content depth but weak on interactivity and accessibility for beginners. The emerging "visual-first, sandbox-driven" learning format — pioneered in adjacent domains by LearnGitBranching (version control), Flexbox Froggy (CSS), and SQL Murder Mystery (databases) — has not yet been applied systematically to networking, representing a clear whitespace opportunity for NetPath.

3.2 Target Market & Competitive Analysis

Addressable Market Segmentation

Segment	Profile	Est. Annual Spend on Networking Ed.
Higher Education	CS/IT students at universities and community colleges	USD 1.2B (global course materials & lab tools)
Professional Self-Learners	Developers, sysadmins, and career changers seeking certs	USD 800M (certification prep courses & labs)
Corporate Training	IT onboarding, helpdesk upskilling, security awareness	USD 3.5B (enterprise technical training budgets)
Secondary Education	High school computing and STEM programmes	USD 400M (emerging, fast-growing segment)

Competitive Landscape

The table below maps key competitors across two dimensions: interactivity (passive video vs. hands-on lab) and accessibility (expert-level vs. beginner-friendly).

Competitor	Strengths	Weaknesses	NetPath Advantage
Cisco Networking Academy (NetAcad)	Deeply respected brand; free tier; Packet Tracer simulator	Steep learning curve; Packet Tracer is heavy desktop software; dated UX	Browser-native, no install; beginner-first pacing; modern UI
CBT Nuggets / Udemy Networking Courses	Breadth of content; video production quality	Passive video format; no interactive sandbox; subscription fatigue	Active learning; immediate feedback; focused scope
GNS3 / EVE-NG	Professional-grade emulation; real IOS images	Requires powerful hardware; extremely complex setup; not for beginners	Zero-config, runs in browser; purpose-built for education not emulation
LearnGitBranching	Excellent UX; visual, interactive; free	Covers only Git; not networking	Same pedagogical model applied to networking
Boson NetSim	Realistic CLI simulation; cert-focused	CLI-only; no visual topology builder; paid, expensive	Visual-first; drag-and-drop; freemium entry point

4. Marketing and Sales Strategy

4.1 Marketing Channels

NetPath will use a layered, cost-efficient channel mix calibrated to each target segment:

Organic / Content Marketing

- Technical blog and YouTube channel publishing networking explainers, topology teardowns, and "build this network" challenge videos — driving inbound discovery from learners already searching for networking tutorials.
- SEO targeting high-intent queries: "how does a router work", "CCNA practice labs free", "interactive networking tutorial", "what is DNS explained visually".
- Open-source release of the validation engine as a standalone npm package, seeding developer community awareness and backlinks.

Community & Social

- Active presence on Reddit (r/networking, r/ccna, r/learnprogramming), Discord server for learner Q&A and challenge sharing, and LinkedIn for corporate segment outreach.
- Partnerships with networking certification influencers and YouTube educators (e.g. Network Chuck, Professor Messer) for sponsored integrations and affiliate programmes.
- "Share your network" feature: completed sandbox designs generate a shareable visual card optimised for Twitter/X and LinkedIn, providing viral organic reach at zero marginal cost.

Academic & Institutional Outreach

- Direct outreach to university and community college CS/IT department heads offering a free institutional tier for classroom use in exchange for student testimonials and case studies.
- Conference presence at SIGCSE (computing education) and EDUCAUSE to build credibility with academic buyers.

4.2 Customer Acquisition

Freemium Funnel

The primary acquisition engine is a generous free tier that delivers genuine value and creates habitual engagement before any monetisation pressure. The conversion funnel works as follows:

Funnel Stage	Mechanism	Target Metric
Awareness	SEO, social sharing, influencer content, word-of-mouth	100K unique visitors / month by Month 18
Activation	First tutorial stage completable with no sign-up required	40% of visitors complete Stage 1
Registration	Prompt to save progress after Stage 3 completion	30% sign-up rate from activated users
Engagement	Email drip campaign: 7-day curriculum nudges, challenge notifications	50% Day-7 retention among registered users
Conversion (B2C)	Paywall at Arc 2 (WAN content) or after 10 free stages	5% free-to-paid conversion rate
Conversion (B2B)	"Invite your team" prompt; demo request flow for HR/L&D contacts	2% of registered users trigger B2B pipeline

Partnerships

- Integration with LMS platforms (Canvas, Moodle, Google Classroom) via LTI connector to unlock institutional purchasing without displacing existing tooling.
- Bundling discussions with CompTIA and EC-Council as a recommended practice tool for Network+ and CEH candidates.
- Corporate pilot programme: 90-day free trials for teams of up to 25, converting to annual seat licences upon positive NPS scores.

4.3 Growth Strategy & Monetisation

Revenue Model

Tier	Price	Inclusions	Target Segment
Free	USD 0 / forever	Arc 1 (LAN curriculum, ~10 stages), 3 sandbox challenges, basic simulation	Students, casual learners, top-of-funnel
Pro	USD 9 / month or USD 79 / year	Full curriculum (Arc 1 + Arc 2, 40+ stages), unlimited sandbox saves, timed challenges, offline mode, certificate of completion	Self-learners, cert candidates
Teams	USD 12 / seat / month (min. 5 seats)	All Pro features, admin dashboard, cohort progress tracking, custom challenge creation, LMS integration, priority support	Corporate training, bootcamps

Tier	Price	Inclusions	Target Segment
Academic	USD 4 / seat / year (min. 30 seats)	All Pro features, instructor view, assignment mode, LTI integration, usage analytics	Universities, secondary schools

18-Month Financial Projections

Milestone	Month	KPI Target
Public beta launch (Arc 1 free)	Month 3	5,000 registered users; NPS > 50
Pro tier launch	Month 6	500 paying Pro subscribers; MRR USD 4,500
Teams tier launch	Month 9	10 corporate pilots; ARR USD 80,000
Arc 2 (WAN content) complete	Month 12	25,000 registered users; 2,000 Pro subs; MRR USD 18,000
Academic partnerships active	Month 15	5 institutional contracts; combined 3,000 seats
Break-even (operating costs)	Month 18	MRR USD 35,000+; total users 75,000+

Long-Term Growth Levers

- Curriculum expansion: advanced modules (BGP, SDN, cloud networking, Kubernetes networking) targeting senior engineers and specialist cert tracks.
- User-generated challenges: a community marketplace where educators and practitioners publish and monetise their own sandbox challenge packs.
- API / embed product: a white-label sandbox widget licensable by universities and certification bodies for embedding in their own learning portals.
- AI-powered hint system: an integrated tutoring assistant that analyses a learner's broken topology and offers targeted, Socratic guidance without giving away the answer directly.

NetPath enters a market ripe for disruption with a clearly differentiated product, a proven pedagogical model, and a capital-efficient go-to-market strategy anchored in organic growth and viral product mechanics. The founding team combines the technical depth to build the product and the educational domain expertise to make it genuinely effective — positioning NetPath to become the definitive first stop for anyone learning how networks work.